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Impedance adaptation for optimal robot–environment interaction

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In this paper, impedance adaptation is investigated for robots interacting with unknown environments. Impedance control is employed for the physical interaction between robots and environments, subject to unknown and uncertain environments dynamics. The unknown environments are described as linear systems with unknown dynamics, based on which the desired impedance model is obtained. A cost function that measures the tracking error and interaction force is defined, and the critical impedance parameters are found to minimise it. Without requiring the information of the environments dynamics, the proposed impedance adaptation is feasible in a large number of applications where robots physically interact with unknown environments. The validity of the proposed method is verified through simulation studies.

Keywords: robot–environment interaction; impedance adaptation; optimal control

1. Introduction

In social applications such as elderly care, health care and human–robot collaboration, environments are typically unknown to robots and there exist uncertainties due to many factors. Therefore, control of interaction between robots and environments is essential and there has been much effort made on this topic. In the literature of interaction control, two approaches are widely studied, i.e., hybrid position/force control (Craig & Raibert, 1979; Huang, Ge, & Lee, 2006; Li, Ge, & Ming, 2007) and impedance control (Hogan, 1985). Compared to hybrid position/force control, impedance control is more feasible in the sense that it does not require the decomposition of two directions. Besides, impedance control is preferred for its better robustness (Colgate & Hogan, 1988).

Under impedance control, robots are governed to be compliant to the interaction force exerted by environments and thus the safety of both robots and environments can be guaranteed. Specifically, imposing a passive impedance model to robots will guarantee the interaction stability if environments are also passive (Colgate & Hogan, 1988). In the early research works of impedance control, a desired passive impedance model is usually prescribed and then the effort is focused on handling the uncertainties in robots dynamics. These works include adaptive impedance control such as Colbaugh, Seraji, and Glass (1991) and Lu and Meng (1991) and learning impedance control such as Cheah and Wang (1998), Wang and Cheah (1998) and

Li, Ge, and Yang (2012). However, in many situations, to impose a passive impedance model to the robot is too conservative, and the environments dynamics can be taken into consideration to obtain desired impedance model (Buerger & Hogan, 2007). Besides, a fixed prescribed impedance model does not suffice in many applications. For example, variable impedance control is necessary in human–robot collaboration (Tsumugiwa, Yokogawa, & Hara, 2001, 2002) and explosive movement (Braun et al., 2012a; Braun, Howard, & Vijayakumar, 2012b). Although the methods discussed in Tsumugiwa et al. (2001, 2002) provide a better control performance in the sense of more efficient human–robot collaboration, the resulted impedance parameters (mass, damping and stiffness) are obtained in a heuristical way and cannot be easily extended to other applications. To cope with this problem, iterative learning has been studied to obtain impedance parameters subject to unknown environments in an analytic way. It has been generally acknowledged that such an ability to improve performance by repeating a task is an important control strategy of the human being (De Roover, Bosgra, & Steinbuch, 2000; Xu, Viswanathan, & Qu, 2000). Pioneered by Arimoto, Kawamura, and Miyazaki (1984a, 1984b), iterative learning control has been widely investigated for robot control. In Cohen and Flash (1991), associative search network is adopted for the impedance learning and the resulted impedance control is applied to a wall-following task. In Yang and Asada (1996), an internal model-based

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impedance learning method is developed and used in a high-speed insertion application. In Tsuji and Morasso (1996), neural networks are employed to update both the impedance parameters and rest position iteratively. Compared to iterative impedance learning discussed above, impedance adaptation is more interesting yet it is more challenging. It is interesting because it does not require the robot to repeat operations to learn the desired impedance parameters. This is important because to make the robot repeat operations may cause inconvenience in many situations. It is challenging because to develop an adaptive scheme usually requires that a certain variable is invariant but this is difficult to satisfy in the case of dynamically changing environment. There has been research effort on impedance adaptation in the literature, although it is less compared to that on impedance learning. In Uemura and Kawamura (2009), stiffness is updated to minimise the actuator torque by taking resonance into consideration. In Stanisić and Fernandez (2012), the switching strategies of impedance parameters are discussed in order to dissipate the system energy and realise a ‘soft’ interaction.

In the development of impedance learning and adaptation, optimisation plays an important role because the control objective of impedance control includes both the force regulation and trajectory tracking and usually it is the compromise of these two objectives. In the case of impedance adaptation, a cost function or a reward function is defined to describe the interaction performance, and impedance parameters are expected to be adjusted to minimise the cost function or maximise the reward function. In Johansson and Spong (1994), the well-known linear quadratic regulator (LQR) is utilised to determine the desired impedance parameters with the environment dynamics known *a priori*. In Matinfar and Hashtrudi-Zaad (2005), impedance parameters are adjusted as the online solutions of the defined LQR problem, instead of fixing the impedance parameters obtained based on LQR as in Johansson and Spong (1994). However, the environment dynamics are also assumed to be known in Matinfar and Hashtrudi-Zaad (2005). Recalling LQR in Kirk (1970), it is difficult to find the solution of the Riccati equation if the linear system under study is unknown. Therefore, when the system dynamics are unknown, the methods proposed in Johansson and Spong (1994) and Matinfar and Hashtrudi-Zaad (2005) are not applicable. To solve the optimal control problems in the case of unknown system dynamics, adaptive dynamic programming (ADP) or reinforcement learning (RL) has been widely studied in the literature (Bertsekas, 1995; Lewis & Vrabie, 2009; Wang, Zhang, & Liu, 2009; Werbos, 1990a, 1990b, 1992, 2009). ADP is constructed based on the idea of how biological system interacts with the surrounding environment. In the scheme of ADP, the control system is defined as an actor or agent, modifying its action based on the feedback information of the environment. The actor or agent is rewarded or punished for a control action which is

evaluated by a critic (Lewis & Vrabie, 2009; Wang et al., 2009). Among all ADP approaches, most recognised discrete ADP algorithms are the heuristic dynamic programming (HDP), action-dependent heuristic dynamic programming (ADHDP) or Q-learning (Watkins, 1989), globalised DHP (GDHP) and dual-heuristic programming (DHP). The common feature of these ADP algorithms is that the design of optimal controller only requires partial information of the system model to be controlled. There are existing works where ADP is adopted for the impedance adaptation of robot arm control. In Kim, Park, Park, and Kang (2010), natural actor-critic algorithm is adopted and the damping and stiffness matrices are updated according to defined reward functions. In Buchli, Stulp, Theodorou, and Schaal (2011), the policy improvement with path integrals (P^2) algorithm is integrated with the RL algorithm to achieve variable impedance control. However, as in Arimoto et al. (1984a, 1984b), Cohen and Flash (1991), Yang and Asada (1996) and Tsuji and Morasso (1996), a learning process is still required in Kim et al. (2010) and Buchli et al. (2011) for the robot to repeat operations to learn the desired impedance parameters. To solve this problem, this paper aims to develop impedance adaptation in the case of unknown environments dynamics. The method to be developed is based on the latest result of ADP in Jiang and Jiang (2012), where the solution of adaptive optimal control is obtained subject to unknown system dynamics. Two general models of environments are considered, one of which includes damping and stiffness, and the other one includes mass, damping and stiffness. These two models are described as linear systems with unknown dynamics. While ADP in Jiang and Jiang (2012) is only for the state regulation, it is further modified to handle the trajectory tracking. The developed impedance adaptation will result in the desired impedance parameters that are able to guarantee the optimal interaction, subject to unknown environments.

Based on the above discussion, we highlight the contributions of this paper as follows:

- (i) Environment dynamics are taken into consideration in the analysis of optimal robot–environment interaction, and they are described as linear systems with unknown dynamics.
- (ii) ADP for systems with unknown dynamics is modified such that trajectory tracking is achievable and the desired impedance model can be obtained.
- (iii) Impedance parameters of robots are obtained subject to unknown environments, which guarantee the optimal robot–environment interaction in the sense of trajectory tracking and force regulation.

The rest of the paper is organised as follows. In Section 2, the dynamics of the robot and environment are described, and impedance control and the objective of this paper are discussed. In Section 3, impedance adaptation

is developed for two general kinds of environments, such that the optimal interaction is achieved subject to unknown environments. In Section 4, the validity of the proposed method is verified through simulation studies. Section 5 concludes this paper.

2. Problem formulation

2.1 System description

The system under study includes a rigid robot arm and an environment, where the end-effector of the robot arm physically interacts with the environment. There is a force sensor at the end-effector of the robot arm which measures the interaction force between the robot arm and the environment, as shown in Figure 1.

Consider the robot kinematics as below

$$\dot{x}(t) = \phi(q), \quad (1)$$

where $x(t) \in \mathbb{R}^n$ and $q \in \mathbb{R}^n$ are positions/oritations in the Cartesian space and joint coordinates in the joint space, respectively. Differentiating Equation (1) with respect to time results in

$$\dot{x}(t) = J(q)\dot{q}, \quad (2)$$

where $J(q) \in \mathbb{R}^{n \times n}$ is the Jacobian matrix.

The robot dynamics in the joint space are given by

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + J^T(q)f, \quad (3)$$

where $M(q) \in \mathbb{R}^{n \times n}$ is the inertia matrix; $C(q, \dot{q})\dot{q} \in \mathbb{R}^n$ denotes the Coriolis and Centrifugal force; $G(q) \in \mathbb{R}^n$ is the gravitational force; $\tau \in \mathbb{R}^n$ is the vector of the con-

trol input; and $f(t) \in \mathbb{R}^n$ denotes the force exerted by the environment.

The other part of the system under study is the environment. Without loss of generality, two kinds of environments are considered in this paper, of which the dynamics are respectively described by the following models:

$$M_E\ddot{x} + C_E\dot{x} + G_Ex = -f \quad (4)$$

$$C_E\dot{x} + G_Ex = -f, \quad (5)$$

where M_E , C_E and G_E are unknown mass, damping and stiffness matrices of the environment models, respectively. Compared to the second model (5), there is a mass matrix in the first model (4). It will be shown that different impedance models of the robot arm are required for different environments (4) and (5) to achieve the optimal interaction.

Remark 1: The above two kinds of environments represent a large range of environments. For examples, model (4) may describe the dynamics of human limb in physical human–robot interaction (Rahman, Ikeura, & Mizutani, 2002), and model (5) may represent the viscoelastic object in robotic manipulation.

Remark 2: M_E , C_E and G_E are assumed to be unknown constant matrices in this paper. While it is valid in many applications, these matrices can be time-varying in some other applications. The latter assumption makes the problem studied in this paper more complicated and it will be further investigated in the future work.

2.2 Impedance control

As discussed in the introduction, impedance control is employed for robots interacting with environments. To implement impedance control, a two-loop control framework is usually adopted, as shown in Figure 2.

In this framework, the outer-loop is dedicated to generate the virtual desired trajectory in the joint space, i.e., q_d . In particular, the desired impedance model in the Cartesian space is given by

$$f = Z(x_d, x_0), \quad (6)$$

where x_d is the desired trajectory, x_0 is the virtual desired trajectory in the Cartesian space and $Z(\cdot)$ is a target impedance

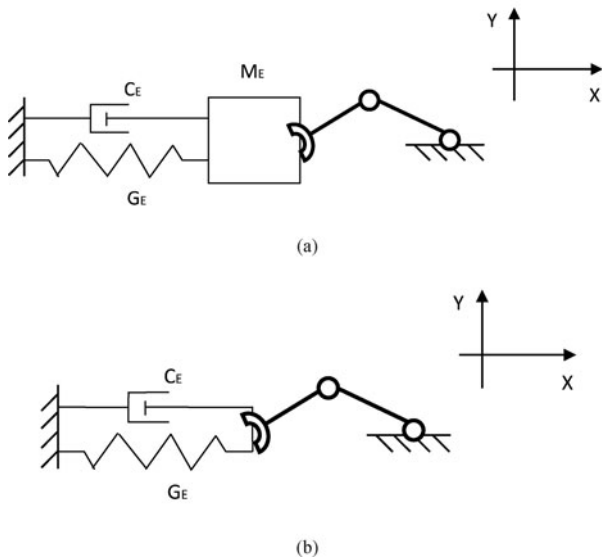


Figure 1. System under study: (a) mass–damping–stiffness environment, and (b) damping–stiffness environment.

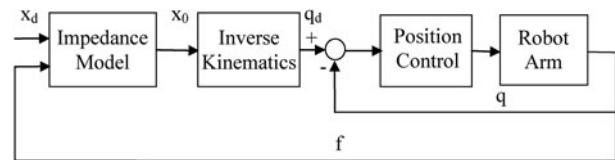


Figure 2. Impedance control diagram.

function to be determined. Then, the virtual desired trajectory in the joint space $q_d = \int_0^t J^{-1}(q(v))\dot{x}_0(v)dv$ is obtained according to the interaction force f and the impedance model (6).

Remark 3: Model (6) is a general impedance model which defines the impedance relationship between interaction force and position. In practical implementations, a typical impedance model is usually given by $M_d\ddot{x}_0 + C_d\dot{x}_0 + G_d(x_0 - x_d) = -f$, where M_d , C_d and G_d are desired inertial, damping and stiffness matrices, respectively. A more simplified model, e.g., stiffness model $G_d(x_0 - x_d) = -f$, may be adopted in a specific situation.

The inner-loop is to guarantee the trajectory tracking, i.e., $\lim_{t \rightarrow \infty} q(t) = q_d(t)$. Trajectory tracking of a robot arm has been extensively studied in the literature (Ge, Lee, & Harris, 1998), and will not be discussed in this paper. For the analysis simplicity, it is assumed that there is an ideal inner-loop position controller such that $q(t) = q_d(t)$ and thus $x(t) = x_0(t)$. In this way, the desired impedance model becomes

$$f = Z(x_d, x). \quad (7)$$

In Equation (7), the impedance function $Z(\cdot)$ is determined such that a certain interaction requirement is satisfied. This is non-trivial considering that the environment dynamics (4) and (5) are unknown. As discussed in the introduction, iterative learning has been studied to cope with this problem, which however requires repetitive motions. In this paper, we aim to achieve the same objective while avoiding the learning process. This is the motivation to develop impedance adaptation in the rest of this paper.

2.3 Preliminary: adaptive optimal control

The adaptive optimal control proposed in Jiang and Jiang (2012) is briefly introduced in this subsection, of which the results will be used for the development of the impedance adaptation.

Consider the following linear system:

$$\dot{\xi} = A\xi + Bu, \quad (8)$$

where $\xi \in \mathbb{R}^p$ is the system state, $u \in \mathbb{R}^r$ is the system input and $A \in \mathbb{R}^{p \times p}$ and $B \in \mathbb{R}^{p \times r}$ are unknown constant matrices.

The following system input:

$$u = -K_k \xi \quad (9)$$

minimises a defined cost function

$$\Gamma = \int_0^\infty [\xi^T Q \xi + u^T R u] dt, \quad (10)$$

where $Q \in \mathbb{R}^{p \times p}$ and $R \in \mathbb{R}^{r \times r}$ are the weights of the state and the input which satisfy $Q = Q^T \geq 0$ and $R = R^T > 0$, and K_k with the iteration number k is a matrix obtained by following the procedures as below.

Step 1: Employ $u = K_0 \xi + v$ as the input on the time interval $[t_0, t_l]$, where K_0 stabilises the system (8) and v is the exploration noise to satisfy the persistent excitation (PE) condition. Compute δ_ξ , I_ξ and I_u until the following rank condition is satisfied:

$$\text{rank}([I_\xi, I_u]) = \frac{p(p+1)}{2} + pr. \quad (11)$$

Step 2: Solve P_k and K_{k+1} according to

$$\begin{bmatrix} \hat{P}_k \\ \text{vec}(K_{k+1}) \end{bmatrix} = (\Theta_k^T \Theta_k)^{-1} \Theta_k^T \Xi_k. \quad (12)$$

Step 3: Let $k + 1 \rightarrow k$ and repeat Step 2 until $P_k - P_{k-1} \leq \epsilon$ where $\epsilon > 0$ is a predefined constant. And K_k in Equation (9) is obtained.

In Step 1, the following definitions are needed:

$$\begin{aligned} \bar{\xi} &= [\xi_1^2, \xi_1 \xi_2, \dots, \xi_1 \xi_p, \xi_2^2, \xi_2 \xi_3, \dots, \xi_{p-1} \xi_p, \xi_p^2]^T \\ \delta_\xi &= [\bar{\xi}(t_1) - \bar{\xi}(t_0), \bar{\xi}(t_2) - \bar{\xi}(t_1), \dots, \bar{\xi}(t_l) - \bar{\xi}(t_{l-1})]^T \\ I_\xi &= \left[\int_{t_0}^{t_1} \xi \otimes \xi dt, \int_{t_1}^{t_2} \xi \otimes \xi dt, \dots, \int_{t_{l-1}}^{t_l} \xi \otimes \xi dt \right]^T \\ I_u &= \left[\int_{t_0}^{t_1} \xi \otimes u dt, \int_{t_1}^{t_2} \xi \otimes u dt, \dots, \int_{t_{l-1}}^{t_l} \xi \otimes u dt \right]^T, \end{aligned} \quad (13)$$

where $\xi_i, i = 1, \dots, p$ are the elements of ξ , l is a positive integer and ' $\otimes$ ' is the Kronecker product.

In Step 2, the following definitions are needed:

$$\begin{aligned} \hat{P}_k &= [P_{1,1}, 2P_{1,2}, \dots, 2P_{1,p}, P_{2,2}, 2P_{2,3}, \dots, \\ &\quad 2P_{p-1,p}, P_{p,p}]^T \\ \Theta_k &= [\delta_\xi, -2I_\xi(I_p \otimes K_k^T R) - 2I_u(I_p \otimes R)] \\ \Xi_k &= -I_\xi \text{vec}(Q_k), \end{aligned} \quad (14)$$

where $P_{i,j}, i = 1, \dots, p, j = 1, \dots, p$ are the elements of P_k , I_p is a p -dimensional unit matrix, ' vec ' is the operator that 'stretches' a matrix to a vector and $Q_k = Q + K_k^T R K_k$.

Remark 4: Note that A and B in Equation (8) are not used in the above procedures so the resulted adaptive optimal control is applicable to the system with unknown dynamics. This is a favourable property compared to the traditional LQR since in many situations the system dynamics are difficult to obtain, if not impossible.

3. Impedance adaptation

This section is dedicated to develop impedance adaptation to achieve optimal interaction subject to unknown environments. Two kinds of environments described by Equations (4) and (5) will be considered in two separate subsections, respectively.

3.1 Mass–damping–stiffness environment

The mass–damping–stiffness environment described by Equation (4) is considered in this subsection. By taking the environment dynamics into consideration, we will develop impedance adaptation such that the following cost function is minimised:

$$\Gamma = \int_0^\infty [\dot{x}^T Q_1 \dot{x} + (x - x_d)^T Q_2 (x - x_d) + f^T R f] dt, \quad (15)$$

where $Q_1 \in \mathbb{R}^{n \times n}$ and $Q_2 \in \mathbb{R}^{n \times n}$ are the weights of the velocity and the trajectory tracking error, respectively, and

$R \in \mathbb{R}^{n \times n}$ has been defined in Equation (8) and is the weight of the interaction force. Besides, $Q_1 = Q_1^T \geq 0$ and $Q_2 = Q_2^T \geq 0$.

Remark 5: Cost functions similar to Equation (15) have been discussed in the related works (Johansson & Spong, 1994; Matinfar & Hashtrudi-Zaad, 2005), which represent the compromise/combination of the force regulation and trajectory tracking and determine the interaction performance. Note that these cost functions are different from that in the traditional LQR problem, where the cost function usually includes the control input and trajectory tracking error. In the traditional LQR problem, the system under study is the robot itself, while the interaction system under study in this paper includes both the robot and the environment.

Comparing the cost function for a general linear system (10) and the defined cost function in this paper (15), some manipulations are needed to make them identical. In particular, we consider

$$\xi = [\dot{x}^T, x^T, z^T]^T, \quad (16)$$

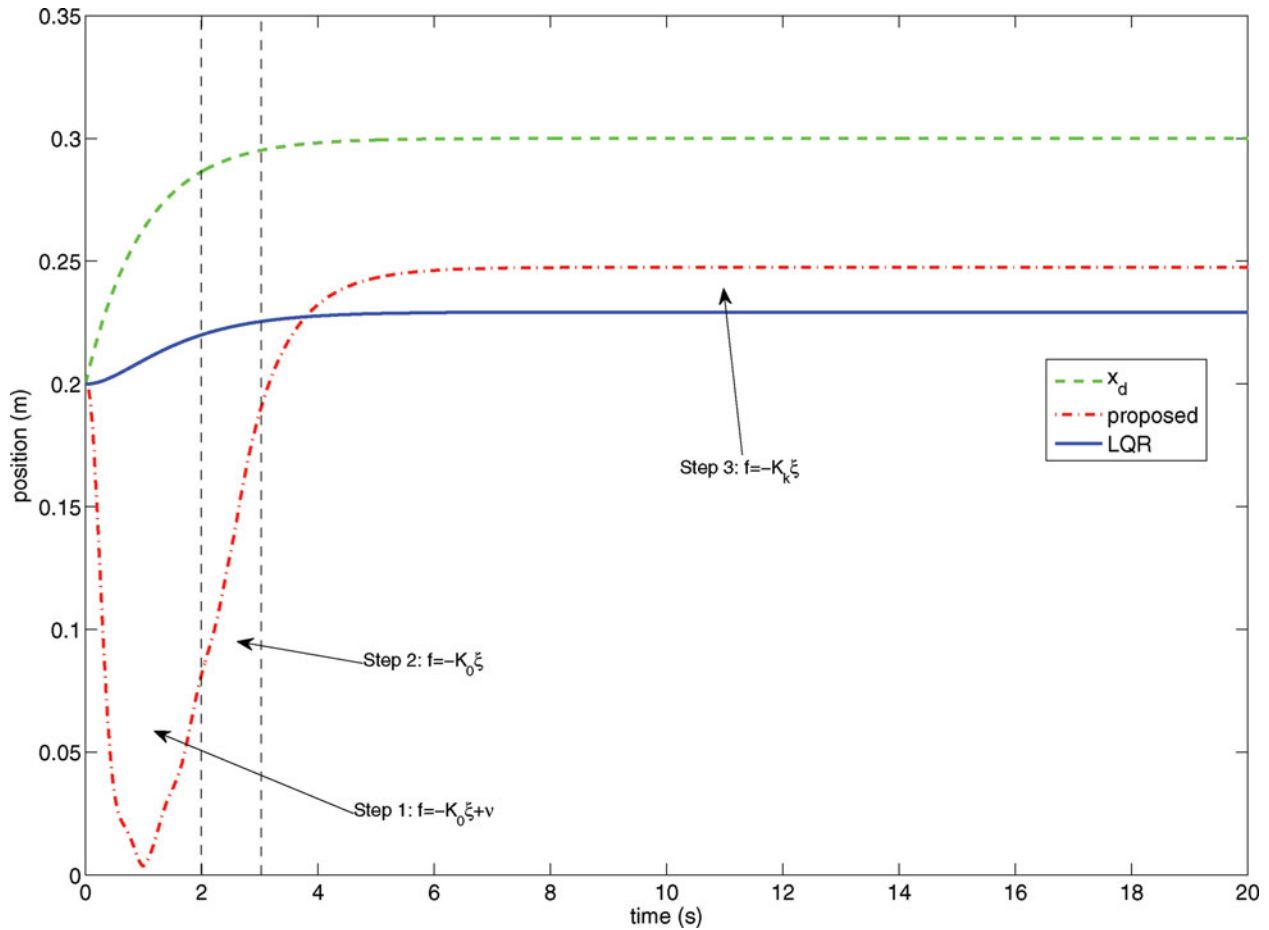


Figure 3. Desired trajectory and actual trajectory, $Q_1 = 1$, $Q_2 = 1$ and $R = 1$.

where $z \in \mathbb{R}^m$ is the state of the following system:

$$\begin{cases} \dot{z} = Uz \\ x_d = Vz, \end{cases} \quad (17)$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times m}$ are two known matrices.

Remark 6: The linear system (17) is to determine the desired trajectory x_d and provides the feasibility to employ the optimal control in trajectory tracking problem. Equation (17) is able to generate a large variety of desired trajectories, e.g., polynomial functions of time of any order.

Then, according to Equations (15) and (17), we have

$$\begin{aligned} \Gamma &= \int_0^\infty \left(\dot{x}^T Q_1 \dot{x} + [x^T x_d^T] \begin{bmatrix} Q_2 & -Q_2 \\ -Q_2 & Q_2 \end{bmatrix} \begin{bmatrix} x \\ x_d \end{bmatrix} \right. \\ &\quad \left. + f^T R f \right) dt \\ &= \int_0^\infty \left(\dot{x}^T Q_1 \dot{x} + [x^T z^T] \begin{bmatrix} Q_2 & -Q_2 V \\ -V^T Q_2 & V^T Q_2 V \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} \right. \\ &\quad \left. + f^T R f \right) dt \\ &= \int_0^\infty (\xi^T Q \xi + f^T R f) dt, \end{aligned} \quad (18)$$

where $Q = \begin{bmatrix} Q_1 & 0 \\ 0 & Q_2' \end{bmatrix}$ with $Q_2' = \begin{bmatrix} Q_2 & -Q_2 V \\ -V^T Q_2 & V^T Q_2 V \end{bmatrix}$.

Considering the defined state (16), we rewrite Equation (4) in the following state-space form

$$\dot{\xi} = A\xi + Bf, \quad (19)$$

where $A = \begin{bmatrix} -M_E^{-1}C_E & -M_E^{-1}G_E & 0 \\ I_n & 0 & 0 \\ 0 & 0 & U \end{bmatrix}$ and $B = \begin{bmatrix} -M_E^{-1} \\ 0 \\ 0 \end{bmatrix}$. Note that A and B include the environment dynamics and they are unknown. This is the main reason to cause the difficulty of determining the desired impedance function $Z(\cdot)$ in Equation (7).

If we take the interaction force f in Equation (19) as the ‘system input’ to the environment dynamics, it can be obtained as follows such that the cost function (15) is minimised

$$f = -K_k \xi, \quad (20)$$

where K_k is obtained according to the procedures described in Section 2.3.

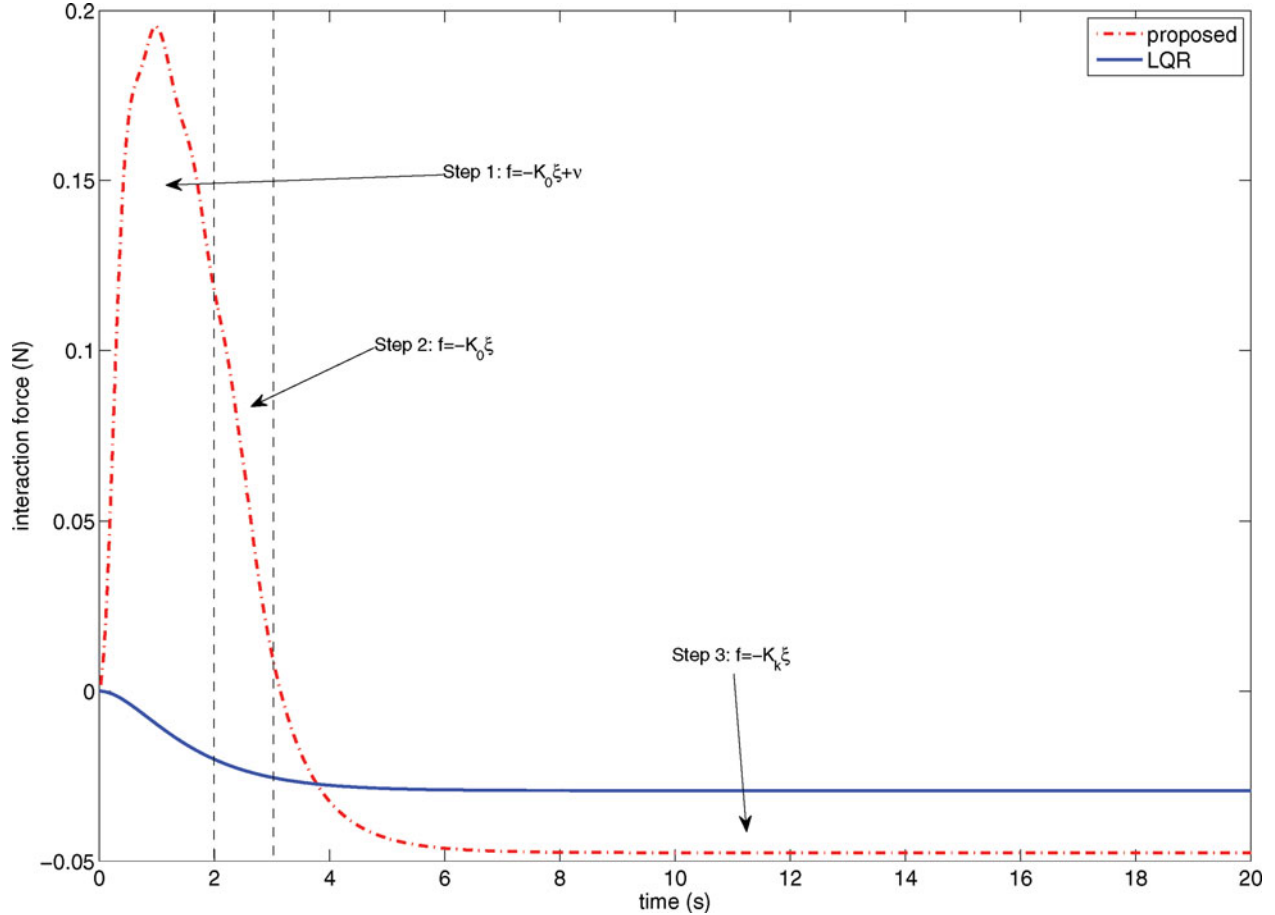


Figure 4. Interaction force, $Q_1 = 1$, $Q_2 = 1$ and $R = 1$.

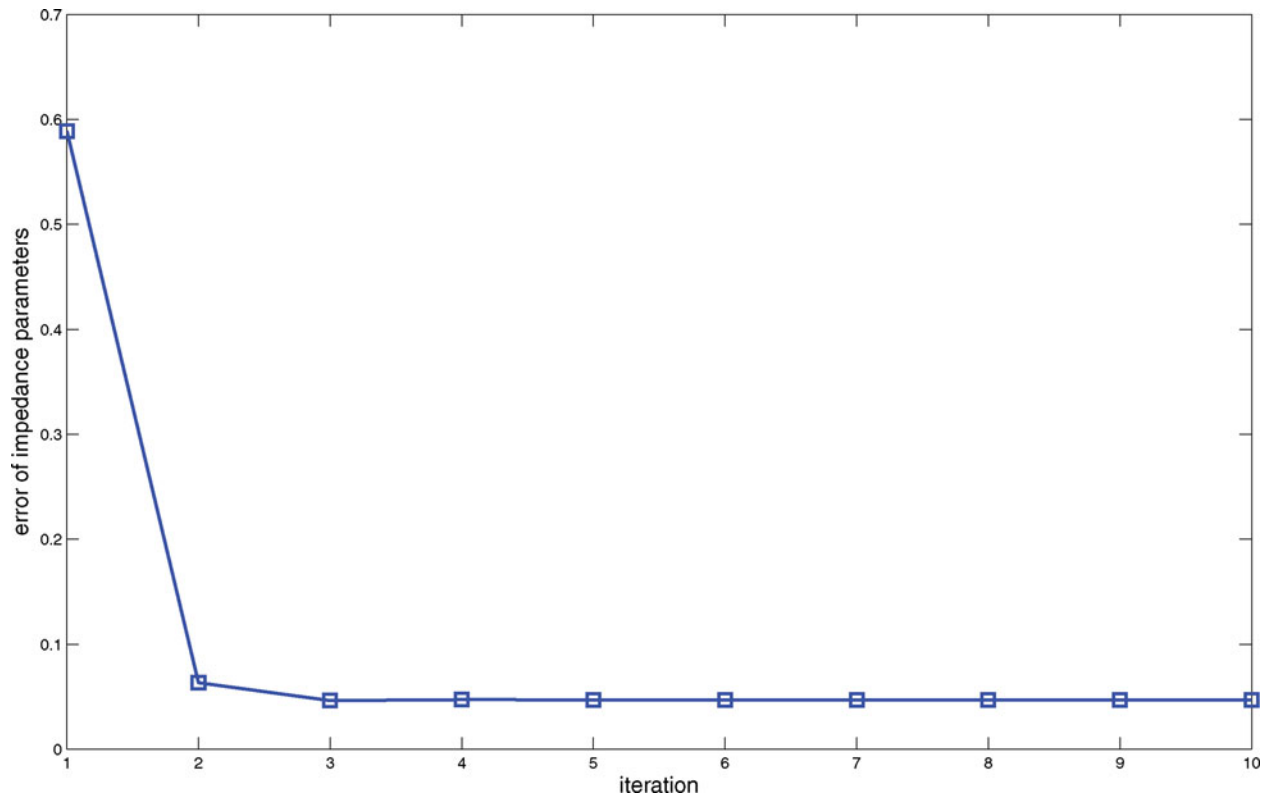


Figure 5. Error of impedance parameters, $Q_1 = 1$, $Q_2 = 1$ and $R = 1$.

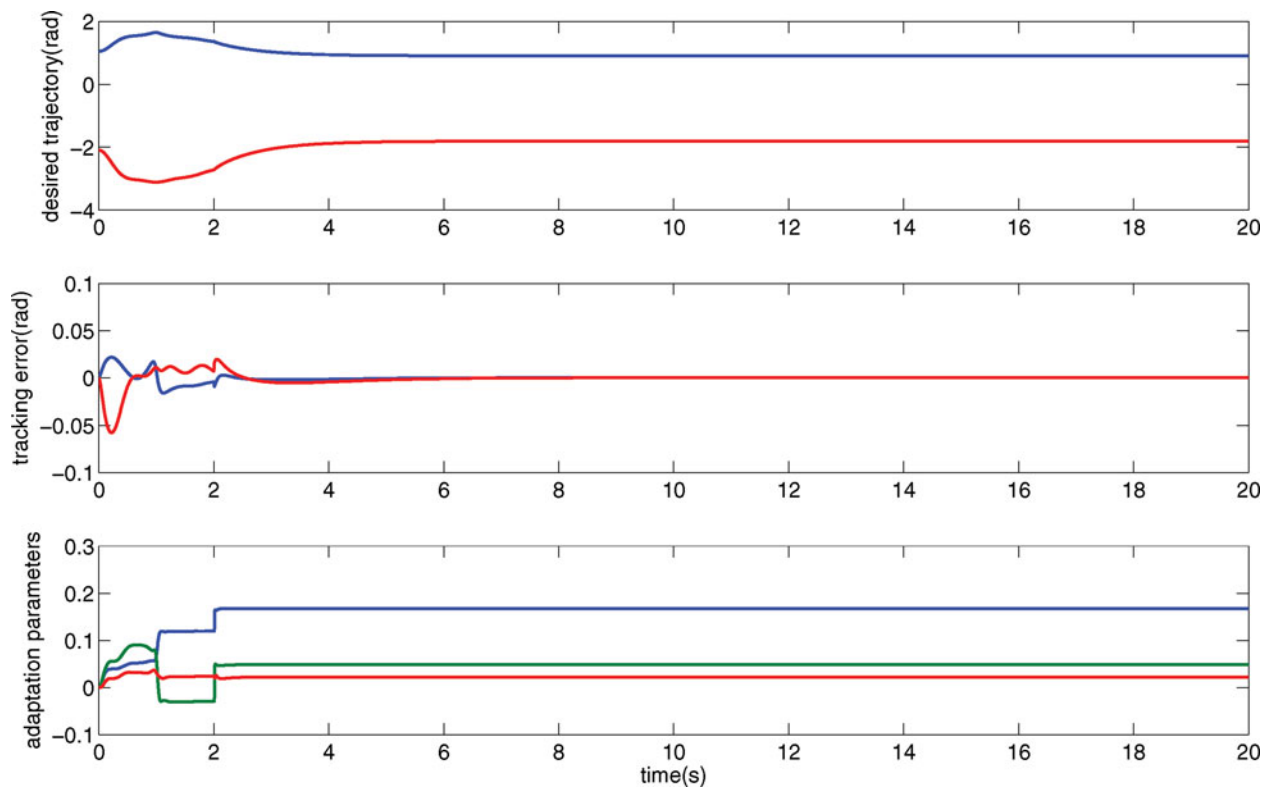


Figure 6. Inner-loop control performance.

Remark 7: For the augmented system, both x and $\dot{x}$ are controllable while the uncontrollable state v is asymptotically stable, so the system is stabilisable.

To understand Equation (20) in the sense of impedance control, we assume that the optimal control has been obtained, which is

$$f = -K\xi = -R^{-1}B^T P\xi, \quad (21)$$

where K is the optimal feedback gain matrix and $P = P^T \in \mathbb{R}^{(2n+m) \times (2n+m)}$ is the solution of the following Riccati equation

$$PA + A^T P - PBR^{-1}B^T P + Q = 0. \quad (22)$$

Denote

$$P = \begin{bmatrix} P_1 & P_2 & P_3 \\ * & * & * \\ * & * & * \end{bmatrix}, \quad (23)$$

where $P_1 \in \mathbb{R}^{n \times n}$, $P_2 \in \mathbb{R}^{n \times n}$, $P_3 \in \mathbb{R}^{n \times m}$ and '*' stand for elements which we do not care about. Substituting Equation (23) into Equation (21) leads to

$$\begin{aligned} f &= -R^{-1}P_1\dot{x} - R^{-1}P_2x - R^{-1}P_3z \\ &= -R^{-1}P_1\dot{x} - R^{-1}P_2x - R^{-1}P_4x_d, \end{aligned} \quad (24)$$

where $P_4 = P_3(V^T V)^{-1}V^T$.

Comparing the above Equation (24) with the desired impedance model (7), the exact impedance function which guarantees the optimal interaction is obtained. Recalling the implementation of impedance control as described in Section 2.2, the virtual desired trajectory in the Cartesian space x is obtained according to Equation (24) with measured f and given x_d , and the inner-position control loop is to guarantee the trajectory tracking in the joint space. In this way, the optimal interaction is achieved and Equation (24) is the resulted impedance function in the presence of unknown environment dynamics.

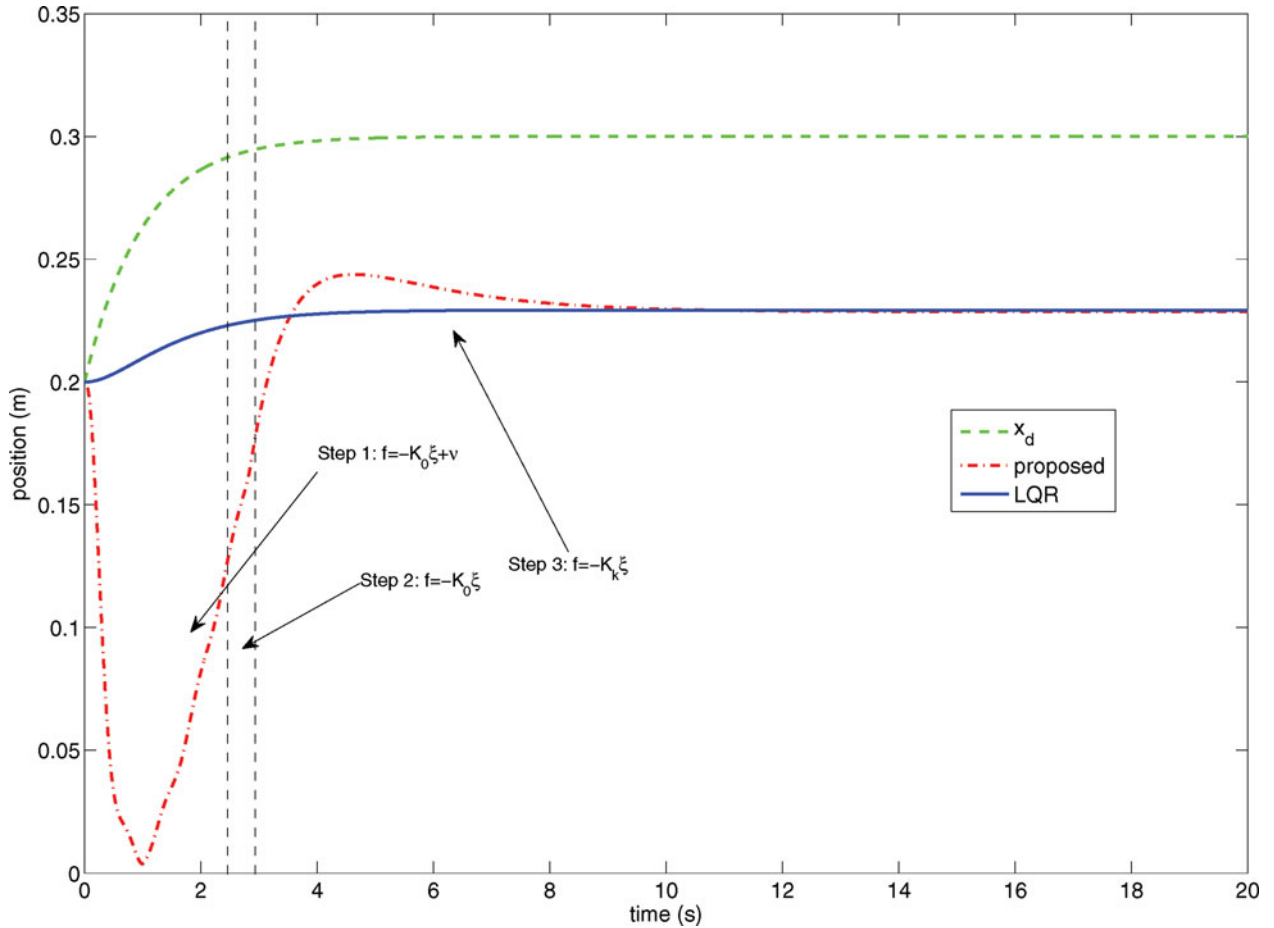


Figure 7. Desired trajectory and actual trajectory, $Q_1 = 1$, $Q_2 = 1$ and $R = 1$.

3.2 Damping–stiffness environment

For the environment without mass, i.e., the damping–stiffness environment described by Equation (5), the resulted impedance adaptation is different and it is discussed in this subsection.

First, we consider to minimise the following cost function:

$$\Gamma' = \int_0^\infty [(x - x_d)^T Q_2 (x - x_d) + f^T R f] dt. \quad (25)$$

Remark 8: Note that the component to penalise the velocity in Equation (15) has disappeared in Equation (25). The reason is that the velocity is not the state of the environment (5), which will be further explained in the following.

Correspondingly, we consider the state $\xi' = [x^T, z^T]^T$, where $z \in \mathbb{R}^m$ has the same meaning as defined in Equation (17). Based on the similar manipulation as in the previous section, Equation (25) can be rewritten as

$$\Gamma' = \int_0^\infty (\xi'^T Q \xi' + f^T R f) dt, \quad (26)$$

where $Q' = \begin{bmatrix} Q_2 & -Q_2 V \\ -V^T Q_2 & V^T Q_2 V \end{bmatrix}$.

Similarly, Equation (5) can be rewritten in the following state-space form

$$\dot{\xi}' = A' \xi' + B' f, \quad (27)$$

where $A' = \begin{bmatrix} -C_E^{-1} G_E & 0 \\ 0 & U \end{bmatrix}$ and $B' = \begin{bmatrix} -C_E^{-1} \\ 0 \end{bmatrix}$.

If we take the interaction force f in Equation (27) as the ‘system input’ to the environment dynamics (5), it can be obtained as follows such that the cost function (25) is minimised

$$f = -K'_k \xi', \quad (28)$$

where K'_k is obtained according to the procedures described in Section 2.3.

To understand Equation (28) in the sense of impedance control, we assume that the optimal control has been obtained, which is

$$f = -K' \xi' = -R^{-1} B'^T P' \xi', \quad (29)$$

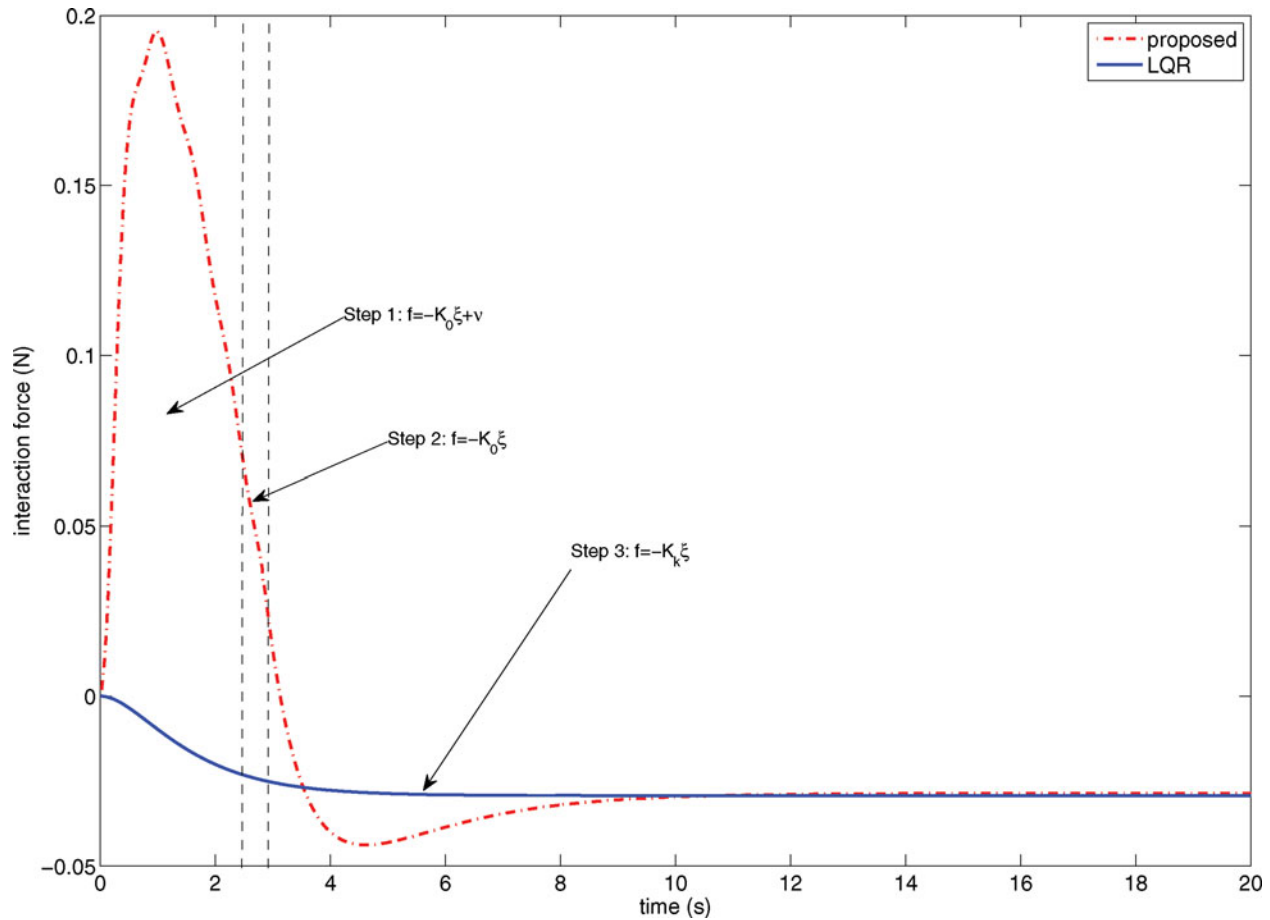


Figure 8. Interaction force, $Q_1 = 1$, $Q_2 = 1$ and $R = 1$.

where K' is the optimal feedback gain matrix and $P' = P'^T \in \mathbb{R}^{(n+m) \times (n+m)}$ is the solution of the following Riccati equation

$$P'A' + A'^T P' - P'B'R^{-1}B'^T P' + Q' = 0. \quad (30)$$

Denote

$$P' = \begin{bmatrix} P'_1 & P'_2 \\ * & * \end{bmatrix}, \quad (31)$$

where $P'_1 \in \mathbb{R}^{n \times n}$ and $P'_2 \in \mathbb{R}^{n \times m}$. Substituting Equation (31) into Equation (29) leads to

$$\begin{aligned} f &= -R^{-1}P'_1 x - R^{-1}P'_2 z \\ &= -R^{-1}P'_1 x - R^{-1}P'_3 x_d, \end{aligned} \quad (32)$$

where $P'_3 = P'_2(V^T V)^{-1}V^T$.

It is found that the resulted control is variable stiffness control, i.e., there is no damping component as in Equation (24). In this sense, we conclude that both the damping and stiffness components are needed in the

impedance adaptation for the optimal interaction with the mass–damping–stiffness environments, and only the stiffness component is needed for the optimal interaction with the damping–stiffness environments. Similarly as in Section 3.1, the desired impedance function $Z(\cdot)$ in Equation (7) is obtained as Equation (32), which guarantees the optimal interaction subject to unknown environment dynamics (5).

Remark 9: The adaptive optimal control proposed in Jiang and Jiang (2012) is limited to linear and time invariant LQR problem, so the methods developed in Sections 3.1 and 3.2 are also limited to linear and time invariant environment dynamics. However, these methods could also be further modified to suit the case where the parameters of the environment dynamics are changing slowly, using a time varying parameter estimation.

4. Simulation study

4.1 Simulation conditions

In this section, we consider a robot arm with two revolute joints physically interacting with two environments, as discussed through this paper and shown in Figure 1(a)

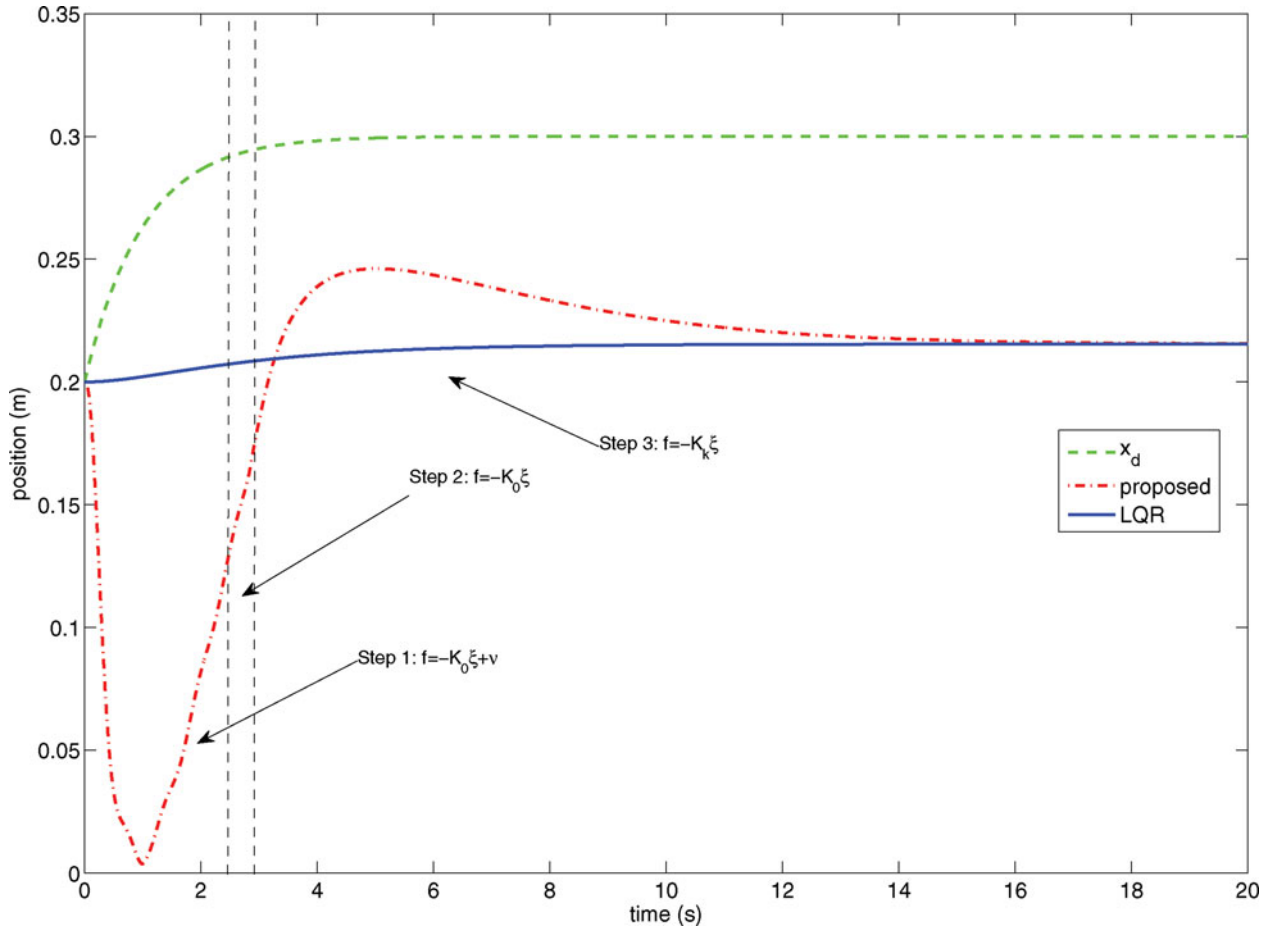


Figure 9. Desired trajectory and actual trajectory, $Q_1 = 1$, $Q_2 = 10$ and $R = 1$.

and 1(b). The simulation is conducted with the Robotics Toolbox (Corke, 1996).

The parameters of the robot arm are: $m_1 = m_2 = 2.0$ kg, $l_1 = l_2 = 0.2$ m, $i_1 = i_2 = 0.027$ kgm², $l_{c1} = l_{c2} = 0.1$ m, where m_j , l_j , i_j , l_{cj} , $j = 1, 2$, represent the mass, the length, the inertia about the z -axis that comes out of the page passing through the centre of mass, and the distance from the previous joint to the centre of mass of link i , respectively.

The initial joint coordinates of the robot arm are $q_1 = -\frac{\pi}{3}$ and $q_2 = \frac{2\pi}{3}$, and thus the initial position in the Cartesian space is $x_d = [0.20]^T$. The desired trajectory in the Cartesian space is determined by Equation (17) with $U = 1$ and $V = 0.3$. It is assumed that the force exerted by the environment is only in X direction and thus the robot arm in Y direction is interaction-free. Nevertheless, the inner-loop position control is designed for both joints. In particular, adaptive control in Slotine and Li (1987) is adopted for the inner-loop position control.

According to the adaptation procedure in Section 3, three steps are included in a single simulation. As discussed in Section 2.3, the exploration noise which meets the persistent excitation condition is required to guarantee

the estimated parameters to converge to the true values. It is well known that the sum of sinusoid waves is a typical swept frequency signal which meets the persistent excitation condition. Therefore, in the simulation, the exploration noise is chosen as $v = \sum_{w=1}^{10} \frac{0.4}{w} \sin(wt)$ and added in the first step. The impedance model for this step is $f = f_0 + v$, where $f_0 = -\dot{x} - x + x_d$ in Section 4.2 and $f_0 = -x + x_d$ in Section 4.3. Impedance adaptation is conducted in the second step, and it stops when the condition $\|P_k\| < 10^{-10}$ satisfies. The initial P_k is $P_0 = 10I_p$, where I_p represents the p dimensional unit matrix. The impedance function for this step is f_0 . The desired impedance model based on the proposed method is obtained at the end of the second step, and it is used in the third step.

Corresponding to Sections 3.1 and 3.2, the following two environments are considered in two subsections: $0.001\ddot{x} + 0.01\dot{x} + (x - 0.2) = -f$ and $0.01\dot{x} + (x - 0.2) = -f$. Note that 0.2 is the initial position of the robot arm. According to Equations (22) and (30), if A and B are known, the optimal solutions of the Riccati equations can be obtained and thus the desired impedance model (24) and (32) can be obtained. It is referred as 'LQR', and compared with the the proposed method in this paper. It is neces-

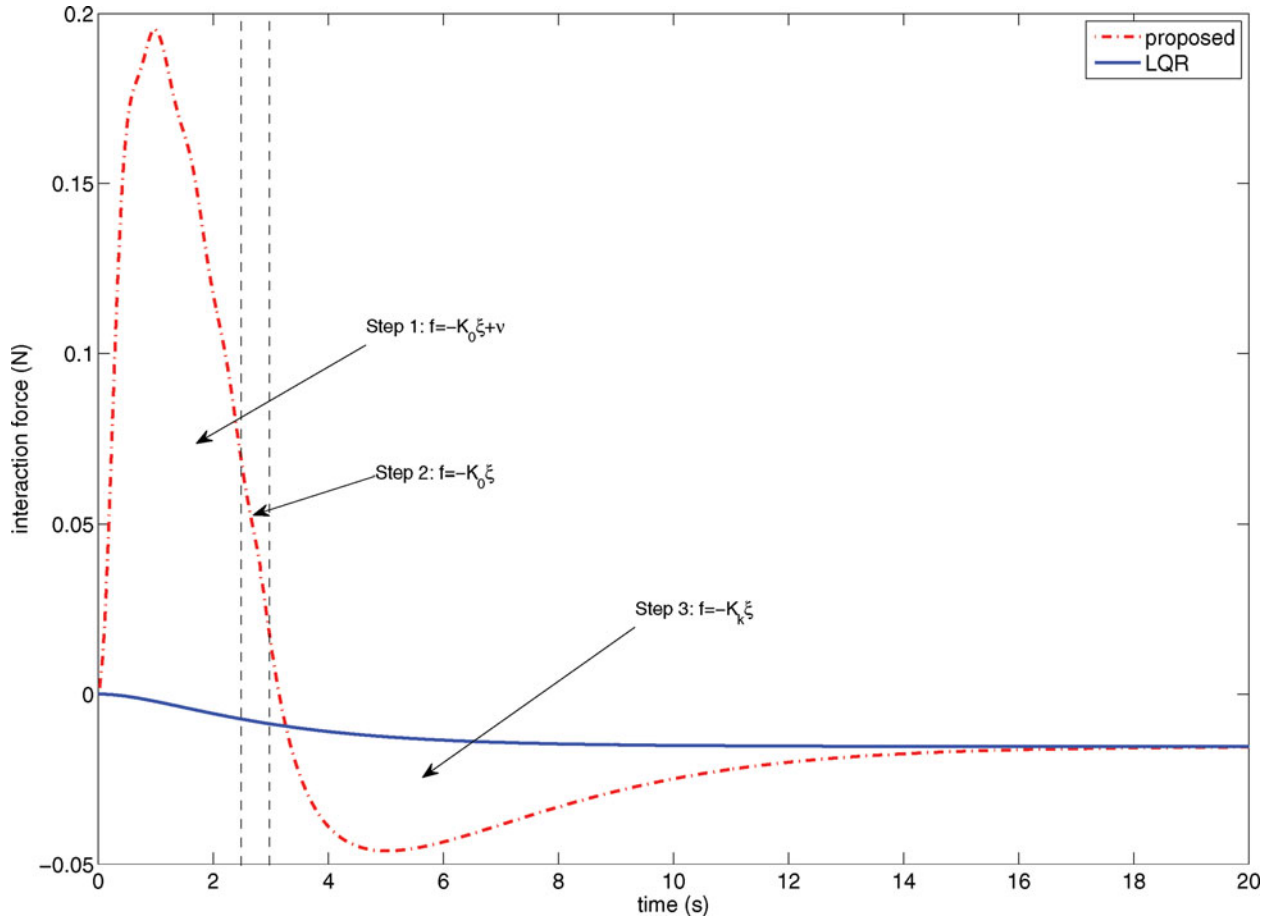


Figure 10. Interaction force, $Q_1 = 1$, $Q_2 = 10$ and $R = 1$.

sary to emphasise that A , B and thus the desired impedance model are only available in the simulation studies for the comparison purpose, and they are usually unknown or need to be estimated in a typical application. This is the motivation of this paper, as already discussed in the introduction.

4.2 Mass–damping–stiffness environment

In this subsection, the mass–damping–stiffness environment $0.001\ddot{x} + 0.01\dot{x} + (x - 0.2) = -f$ is considered. In the first case, the weights in Equation (15) are given by $Q_1=1$, $Q_2=1$ and $R=1$. The desired impedance model (24) is $f = -0.99\dot{x} - 0.41x + 0.04x_d$ based on known A and B . The simulation results are shown in Figures 3–6. In Figure 3, the position of the robot arm in the Cartesian space is shown. In the first two seconds, there is a large position error between LQR and the proposed method. This is because there is exploration noise and the initial impedance model $f = -\dot{x} - x + x_d + v$ is not the desired model. The second step takes a very short time and the desired impedance function converges very quickly. More details about the convergence performance can be found in Figure 5, where

the error of impedance parameters with respect to iteration number is shown. This error is defined as $\|K_k - K\|$, and it decreases to around 0.05 when the adaptation stops at the 5th iteration (each iteration takes 0.1 s). The desired impedance model is obtained as $f = -0.98\dot{x} - 0.38x + 0.06x_d$ with the proposed method, and it is used until the end of the simulation. It is found that the obtained impedance model is very near to but not exactly the same as the desired one under LQR, i.e., $f = -0.99\dot{x} - 0.41x + 0.04x_d$. As a result, the position in Figure 3 is near to the position under LQR but there is still a small error. This is caused by the adaptation process in the inner-loop, and it is illustrated by Figure 6 where the control performance of the inner-loop is shown. Particularly, although zero tracking error in the inner-loop is achieved after about 2s, the tracking error exists at the beginning of the simulation. Therefore, the dynamics of the robot arm are actually not exactly governed by the given impedance model.

To further investigate the effect of inner-loop performance on the impedance adaptation, an additional simulation study is conducted. Instead of starting from the beginning of the simulation, the impedance adaptation

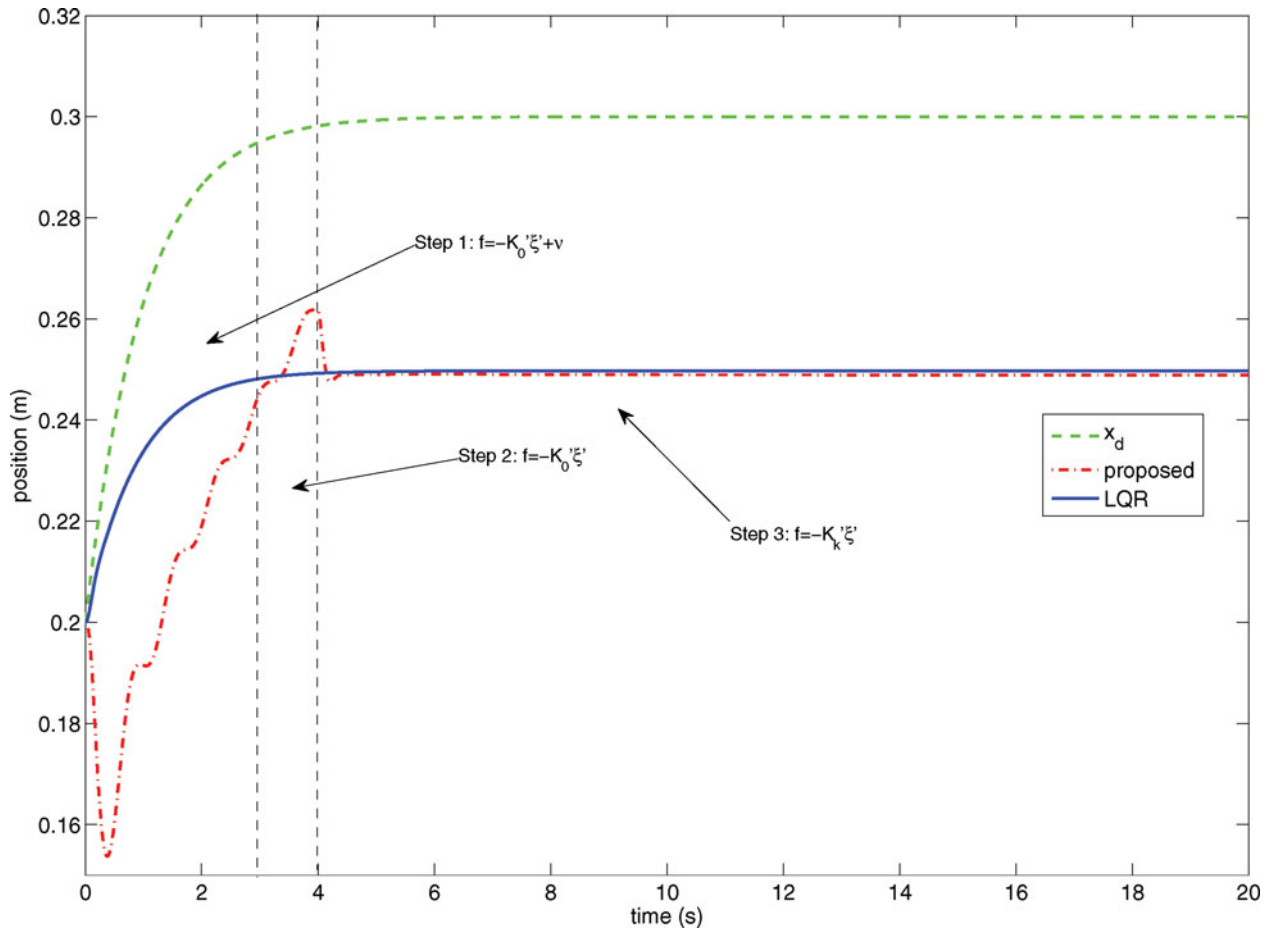


Figure 11. Desired trajectory and actual trajectory, $Q_2 = 1$ and $R = 1$.

starts at 2s when the inner-loop tracking error is near to zero. Simulation results are shown in Figures 7 and 8. The desired impedance model is obtained as $f = -0.98\dot{x} - 0.40x + 0.04x_d$ with the proposed method, which shows that the perfect adaptation can be achieved when a perfect tracking is guaranteed throughout the parameter adaptation period. By comparing the simulation results in Figures 3, 4, 7 and 8, we conclude that a good inner-loop performance during the parameter adaptation period is crucial to guarantee a perfect impedance adaptation. When the good control performance of the inner-loop is not achieved during the parameter adaptation period, the adapted parameters cannot converge to the exact values. Conversely, when a perfect inner-loop control performance is guaranteed during the adaptation process, the perfect impedance adaptation can be realised.

To further verify the effectiveness of the proposed impedance adaptation, another cost function is chosen in the second case. Particularly, the weights in Equation (15) are given by $Q_1 = 1$, $Q_2 = 10$ and $R = 1$. Compared to that in the first case, the weight of the velocity is larger so it is expected that the system response is slower. Similarly, the desired impedance model (24) is obtained as $f = -3.15\dot{x} - 0.41x + 0.02x_d$ based on known A and B .

The simulation results in this case are given in Figures 9 and 10, and the impedance model obtained with the proposed method is $f = -3.14\dot{x} - 0.40x + 0.02x_d$. While the position in Figure 3 converges to around 0.25 m in 6s, the position in Figure 9 converges to around 0.22 m in 10s. Similarly, the interaction force in Figure 4 converges to around -0.05 N in 8s, and in Figure 10 it converges to around -0.02 N in 16s. The above results are coherent with the expectation and verify the validity of the propose method. Similarly as in LQR, different Q_1 , Q_2 and R can be chosen to realise different interaction performances, e.g., either 'softer' interaction or more accurate trajectory tracking.

4.3 Damping–stiffness environment

The damping–stiffness environment $0.01\dot{x} + (x - 0.2) = -f$ is considered in this subsection. The simulation conditions are the same as in the previous subsection, which are described in Section 4.1. The weights in Equation (25) are given by $Q_2 = 1$ and $R = 1$ and the desired impedance model (32) is obtained as $f = -0.41x + 0.70x_d$ based on known A' and B' . Similar results as in the previous subsection are obtained. In particular, the impedance model obtained with

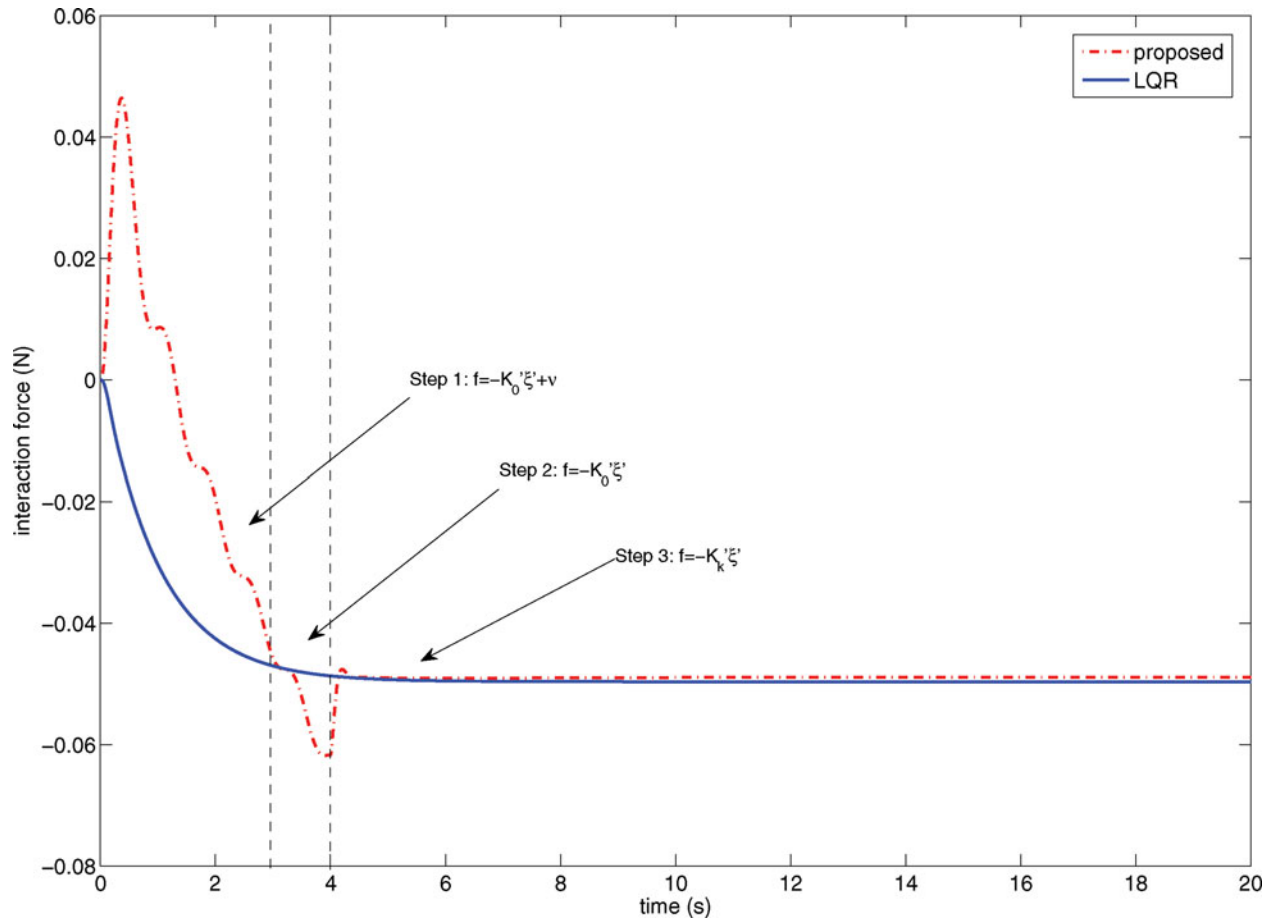


Figure 12. Interaction force, $Q_2 = 1$ and $R = 1$.

the proposed method is $f = -0.40x + 0.71x_d$. The optimal interaction performance is achieved as shown in [Figures 11](#) and [12](#)

4.4 Discussion

In summary, it has been shown that the proposed method can be adopted to obtain a desired impedance model based on a given cost function which determines an optimal interaction performance. Compared to impedance learning developed in the literature such as Kim et al. (2010) and Buchli et al. (2011), the proposed method does not require the repetitive learning process and thus provides certain convenience. The optimal impedance adaptation can be achieved if the ‘perfect’ tracking can be guaranteed in the inner-loop. As discussed in Remark 9, the proposed method in Jiang and Jiang (2012) may not be applicable to the scenario where the environment is changing rapidly. In the future work, we will investigate how to derive an impedance adaptation in face of dynamically changing environments. Besides, some issues in real-world implementations might not be well described in the simulation studies, so future work will also be dedicated to incorporate the proposed impedance adaptation with a real robot arm and further investigate the issues in real-world implementations (e.g., the effect of exploration noise, environment model uncertainties and time delay) of the proposed impedance adaptation. Furthermore, as discussed in Braun et al. (2012b), it is non-trivial to find a proper cost function in many situations, which will be also one of the future works that we will focus on.

5. Conclusion

In this paper, impedance adaptation has been developed to obtain the desired impedance parameters such that the optimal interaction is realised subject to unknown environments. The dynamics of the unknown environments have been investigated and the interaction requirement has been described by minimising a certain cost function which includes trajectory tracking and force regulation. Adaptive optimal control for unknown linear system has been employed as the fundamental of the proposed method. The validity of the proposed method has been verified through simulation studies.

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